Randomized Algorithms

Why randomized algorithms may be interesting?

Equalitarianism between inputs!

No more good or bad inputs, only good or bad luck

Randomized Algorithms

<u>Definition</u>:

There are two kinds of randomized algorithms:

1. a *Las Vegas* algorithm returns a correct answer with probability 1, and we bound the **<u>expected</u>** number of steps it performs.
2. a *Monte Carlo* algorithm returns a correct answer with probability less than 1, and we bound the **<u>error probability</u>** and the **<u>worst case</u>** number of steps it performs.

Randomized Algorithms

What is the expectation?

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What is an error probability?

Basic Probability

Definitions:

1. Ω is a space (set).
2. $A \subseteq \Omega$ is an event.
3. $P: \Omega \longmapsto [0,1]$ is a probability function if $\sum_{\omega \in \Omega} P(\omega) = 1$.
4. A probability space is a pair (Ω, P).
5. A random variable is a function $X: \Omega \longmapsto \mathbb{R}$.

   We can define the sum of random variables as:

$$(X + Y)(\omega) = X(\omega) + Y(\omega).$$

Basic Probability

Example:

Tossing a coin:

$\Omega = \{0,1\}$ is the space

$A = \{0\} \subseteq \Omega$ is a possible event

$P: \Omega \longmapsto [0,1]$ where P(0)=1/2 and P(1)=1/2 is a probability function. Note that $\sum_{\omega \in \Omega} P(\omega) = 1$.

Consider the following game:

If the coin toss result is 1, I get 1$, otherwise I get 0$.

X can be a random variable representing my game profit.

Basic Probability

Definitions (cont.):

6. For a discrete variable X, the expectation of X is:

$$E[X] = \sum_{\omega \in \Omega} X(\omega) \cdot P(\omega)$$

In the last slide game example the expected profit value is:

$$E[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$$

The expectation is a linear operator:

- $E[X+Y] = E[X] + E[Y]$
- $E[c \cdot X] = c \cdot E[X]$

Randomized QuickSort

Randomized QuickSort Scheme:

Input: a set S of size n.

Output: S ordered from minimal to maximal element.

1) Choose *at random* an element $y \in S$ as a pivot.
2) By comparison find the following sets:

$$S_1 = \{x \in S | x < y\},$$
$$S_2 = \{x \in S | x > y\}$$

3) Recursively sort S_1 and S_2
4) Return $\langle sorted\ S_1, y, sorted\ S_2 \rangle$

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How to compute step 1?

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With $\lceil \log n \rceil$ random bits we can get a number in the range $0 - (n-1)$

Randomized QuickSort

Example: $S = \{7,5,2,8,9,1,4,6\}$, n=8

Pos: 0 1 2 3 4 5 6 7

Randomized QuickSort

Example: $S = \{7,5,2,8,9,1,4,6\}$, n=8

Pos: 0 1 2 3 4 5 6 7

Random bits: 100

y=9

$S_1 = \{7,5,2,8,1,4,6\}$

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Random bits: 010

y=2

$S_1 = \{1\}$ $S_2 = \{7,5,8,4,6\}$

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Random bits: 001

y=5

$S_1 = \{4\}$ $S_2 = \{7,8,6\}$

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Pos: 0 1 2

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Pos: 0 1 2 3 4 5 6

Random bits: 010

y=2

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$S_2 = \{7,5,8,4,6\}$ $S_2 = \{4,5,6,7,8\}$

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Randomized QuickSort

Complexity depends on the random bits

$= \{7,5,2,8,9,1,4,6\}$, n=8 $\quad S = \{1,2,4,5,6,7,8,9\}$

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Randomized QuickSort

In fact,

The Complexity=Number of Comparisons

which is a random variable X!

How to compute the expectation of X?

Randomized QuickSort - Analysis

We will compute the expected number of comparisons that the algorithm performs for any input.
For $1 \leq i \leq n$ denote by S^i the element in S that is in the i-th position in the sorted list.

For example: If $S = \{7,5,2,8,9,1,4,6\}$ then after the sorting we have:

$$S = \{1,2,4,5,6,7,8,9\}$$

so,

$$S^1 = 1,\ S^2 = 2,\ S^3 = 4,\ S^4 = 5,\ S^5 = 6,\ S^6 = 7,\ S^7 = 8,\ S^8 = 9$$

Randomized QuickSort - Analysis

We will look at a specific running of the algorithm and define the following set of random variables values according to the execution.

For $1 \leq i, j \leq n$,

$$X_{i,j} = \begin{cases} 1, & if\ S^i\ is\ compared\ to\ S^j \\ 0, & otherwise \end{cases}$$

Randomized QuickSort - Analysis

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$X_{1,2} = 1$
$X_{1,3} = 0$
$X_{1,4} = 0$
$X_{1,5} = 0$
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$X_{1,7} = 0$
$X_{1,8} = 1$

Randomized QuickSort - Analysis

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$X_{7,8} = 1$

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^{n}\sum_{j>i} X_{ij}\right] = \sum_{i=1}^{n}\sum_{j>i} E[X_{ij}] = \sum_{i=1}^{n}\sum_{j>i} P[X_{ij} = 1]$$

$$E[X_{i,j}] = 0 \cdot P\left[X_{i,j} = 0\right] + 1 \cdot P\left[X_{i,j} = 1\right]$$

Randomized QuickSort - Analysis

Observe the first time that a pivot $y = S^k$ is chosen where $i \leq k \leq j$

Claim 1: $X_{ij} = 1$ if and only if $k \in \{i,j\}$

Proof: Assume $X_{ij} = 1$.

Then, S^i is compared during run time with S^j. Note that until $y = S^k$ is chosen where $i \leq k \leq j$, S^i and S^j both belong to same set, either to S_1 or to S_2 and are only compared to the pivots but not to each other.

Now, if it holds that $i < k < j$, then at this phase S^i will belong to S_1 and S^j will belong to S_2 and from now on they will never be compared (because elements are only compared to pivots chosen from the set they belong to). Therefore, it must be that: $k \in \{i, j\}$.

Randomized QuickSort - Analysis

Observe the first time that a pivot $y = S^k$ is chosen where $i \leq k \leq j$

Proof: (cont.)
Assume $k \in \{i,j\}$.
Again, note that until $y = S^k$ is chosen where $i \leq k \leq j$, S^i and S^j both belong to same set, either to S_1 or to S_2 and are only compared to the pivots but not to each other.

Now, if either $y = S^i$ or $y = S^j$ is the pivot, the other is an element in this set and, therefore, must be compared to the pivot. Thus, we have that: $X_{ij} = 1$.

Randomized QuickSort - Analysis

Observe the first time that a pivot $y = S^k$ is chosen where $i \le k \le j$

Claim 2: $p_{ij} = P[X_{ij} = 1] = \frac{2}{j-i+1}$

Proof:

There are only $j - i + 1$ possible values that affect the probability that S^i and S^j are compared (because in the other cases they still belong to the same set), by claim 1.

By claim 1, the first time that a pivot $y = S^k$ is chosen where $i \le k \le j$ only two choices will cause that $X_{ij} = 1$.

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^{n} \sum\nolimits_{j>i} X_{ij}\right] = \sum_{i=1}^{n} \sum\nolimits_{j>i} E[X_{ij}] = \sum_{i=1}^{n} \sum\nolimits_{j>i} P[X_{ij} = 1]$$

$$= \sum_{i=1}^{n} \sum\nolimits_{j>i} p_{ij} =$$

$p_{i,j} = \frac{2}{j-i+1}$

by claim 2!

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

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$$= \sum_{i=1}^{n} \sum_{j>i} p_{ij} = \sum_{i=1}^{n} \sum_{j>i} \frac{2}{j-i+1} = \sum_{i=1}^{n} \sum_{k=1}^{n-i} \frac{2}{k+1}$$

$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^{n}\sum_{j>i} X_{ij}\right] = \sum_{i=1}^{n}\sum_{j>i} E[X_{ij}] = \sum_{i=1}^{n}\sum_{j>i} P[X_{ij} = 1]$$

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$$2\sum_{i=1}^{n}\sum_{k=1}^{n} \frac{1}{k+1}$$

$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^{n} \sum_{j>i} X_{ij}\right] = \sum_{i=1}^{n} \sum_{j>i} E[X_{ij}] = \sum_{i=1}^{n} \sum_{j>i} P[X_{ij} = 1]$$

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$$2\sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k+1} \leq 2 \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{1}{k} =$$

$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!

Randomized QuickSort - Analysis

Analysis of expectation of comparisons number :

$$E[X] = E\left[\sum_{i=1}^{n}\sum_{j>i} X_{ij}\right] = \sum_{i=1}^{n}\sum_{j>i} E[X_{ij}] = \sum_{i=1}^{n}\sum_{j>i} P[X_{ij}=1]$$

$$= \sum_{i=1}^{n}\sum_{j>i} p_{ij} = \sum_{i=1}^{n}\sum_{j>i}\frac{2}{j-i+1} = \sum_{i=1}^{n}\sum_{k=1}^{n-i}\frac{2}{k+1} \le$$

$$2\sum_{i=1}^{n}\sum_{k=1}^{n}\frac{1}{k+1} \le 2\sum_{i=1}^{n}\sum_{k=1}^{n}\frac{1}{k} = 2\sum_{i=1}^{n}\ln(n) = O(n\log n)$$

$$p_{i,j} = \frac{2}{j-i+1}$$

by claim 2!